Null-Constrained Geometry of Heterogeneous Dynamical Systems: A Statistical and Topological Audit of Transition-Field Structure

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We investigate whether latent geometric structure extracted from heterogeneous nonlinear dynamical systems reflects genuine dynamical organization or statistical artifact. A 211-system ensemble spanning 13 canonical dynamical families was projected into a four-dimensional composite manifold derived from 17 normalized dynamical descriptors. Seven null-model classes and seven falsification criteria were applied, including covariance-preserving Gaussian controls, spectral randomization, causal destruction hierarchies, adversarial low-rank embeddings, and representation robustness analysis.

The resulting geometry exhibits partial continuity under parameter variation, robust topological segmentation through flow-based ridge analysis, and selective collapse under row-shuffle destruction. However, covariance-preserving nulls reproduce much of the manifold dimensionality, indicating that second-order feature statistics strongly constrain the observed geometry.

The strongest evidence for genuine dynamical structure is the selective collapse under system-level covariance destruction and the failure of adversarial low-rank nulls to reproduce the metric fingerprint. The strongest evidence against uniqueness is that several statistical nulls reproduce participation-ratio structure nearly exactly.

These results support a constrained interpretation: the observed manifold reflects a mixture of dynamical organization and statistical structure, rather than a universal emergence ontology.

I. INTRODUCTION

Many nonlinear systems exhibit low-dimensional structure despite high-dimensional microscopic dynamics. Classical examples include Lorenz attractors [1], reaction–diffusion systems [2], Ising criticality, coupled oscillators [3], and stochastic branching processes [4].

Recent work in manifold learning [5–8], information geometry [9, 10], and nonlinear dynamics [11, 12] has motivated attempts to construct latent coordinate systems capable of comparing heterogeneous systems through shared geometric structure. However, latent manifolds are notoriously vulnerable to statistical artifacts [13], embedding distortions [14], and covariance-induced topology [15].

The central question of this work is therefore not whether a latent geometry can be constructed, but whether the resulting geometry survives aggressive null-model falsification [16, 17].

II. METHODS

A. System Ensemble

The ensemble contains 211 systems across 13 dynamical families: Lorenz [1], Rössler [18], Hénon [19], logistic maps [4], Kuramoto oscillators, Ising systems, reaction–diffusion models [2], branching processes, Boolean networks, and stochastic processes.

B. Feature Construction

Each system is represented by a 17-dimensional normalized descriptor vector including:

- principal-component structure [20],
- temporal correlation [21],
- replay stability [22],
- effective rank [23, 24],
- ablation sensitivity,
- modal contribution metrics.

The latent coordinates (C, F, A, R) are linear composites of z-scored features [9].

III. RESULTS

A. Global Manifold Geometry

Figure 1 shows the four-dimensional manifold projected into its principal geometric embedding.

The geometry exhibits partial continuity under parametric variation [25] while preserving substantial family-level separation.

B. Flow Topology and Ridge Structure

Density-flow coupling is strongly negative ($r = -0.95$), indicating that high-flow systems preferentially occupy sparse transition regions [26, 27].

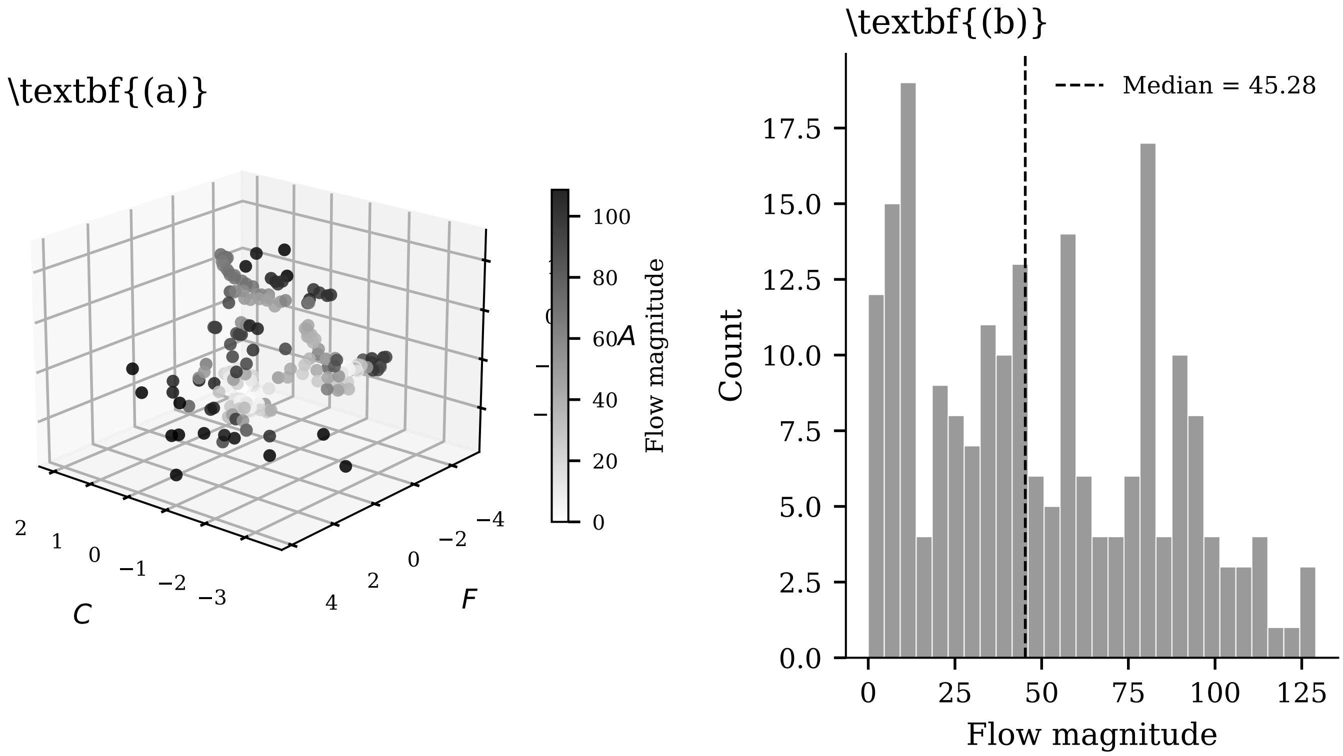

FIG. 1: Global manifold structure in latent coordinate space. Points correspond to individual dynamical systems and are colored by flow magnitude. High-flow systems occupy low-density transition regions connecting otherwise separated families.

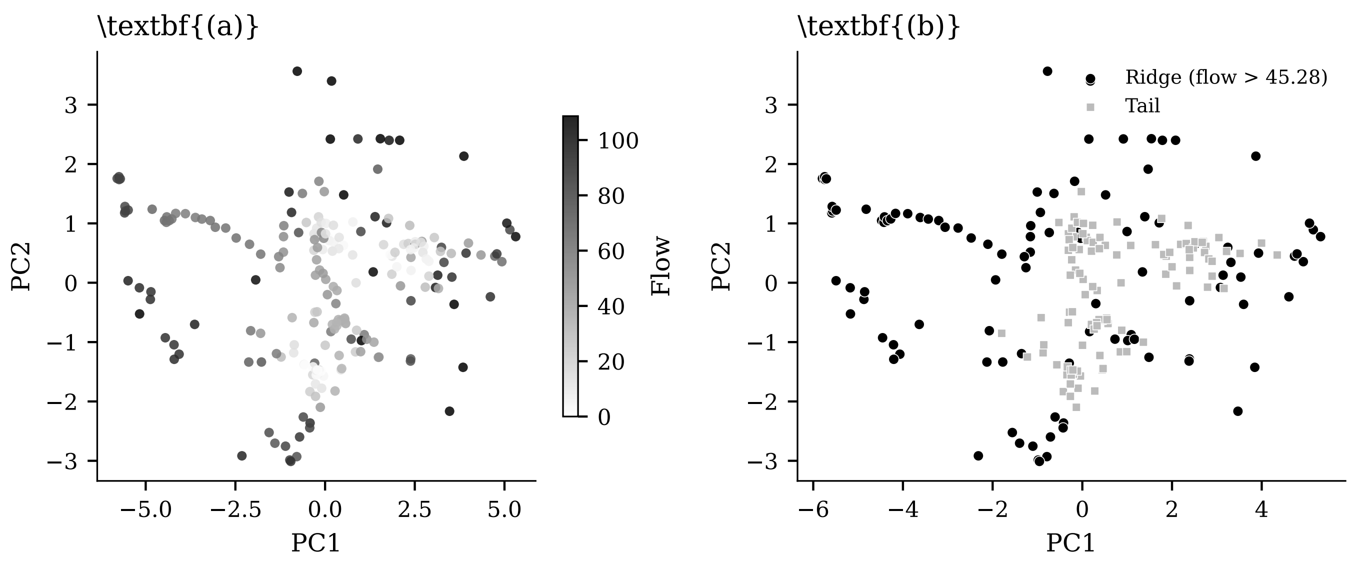

FIG. 2: Flow topology and ridge segmentation analysis. Local density minima coincide with elevated flow magnitude and bifurcation-sensitive regions [12].

C. Null-Model Survival Analysis

Covariance-preserving Gaussian nulls reproduce the participation ratio within 0.3%, demonstrating that second-order statistics [14] account for a substantial fraction of the geometry.

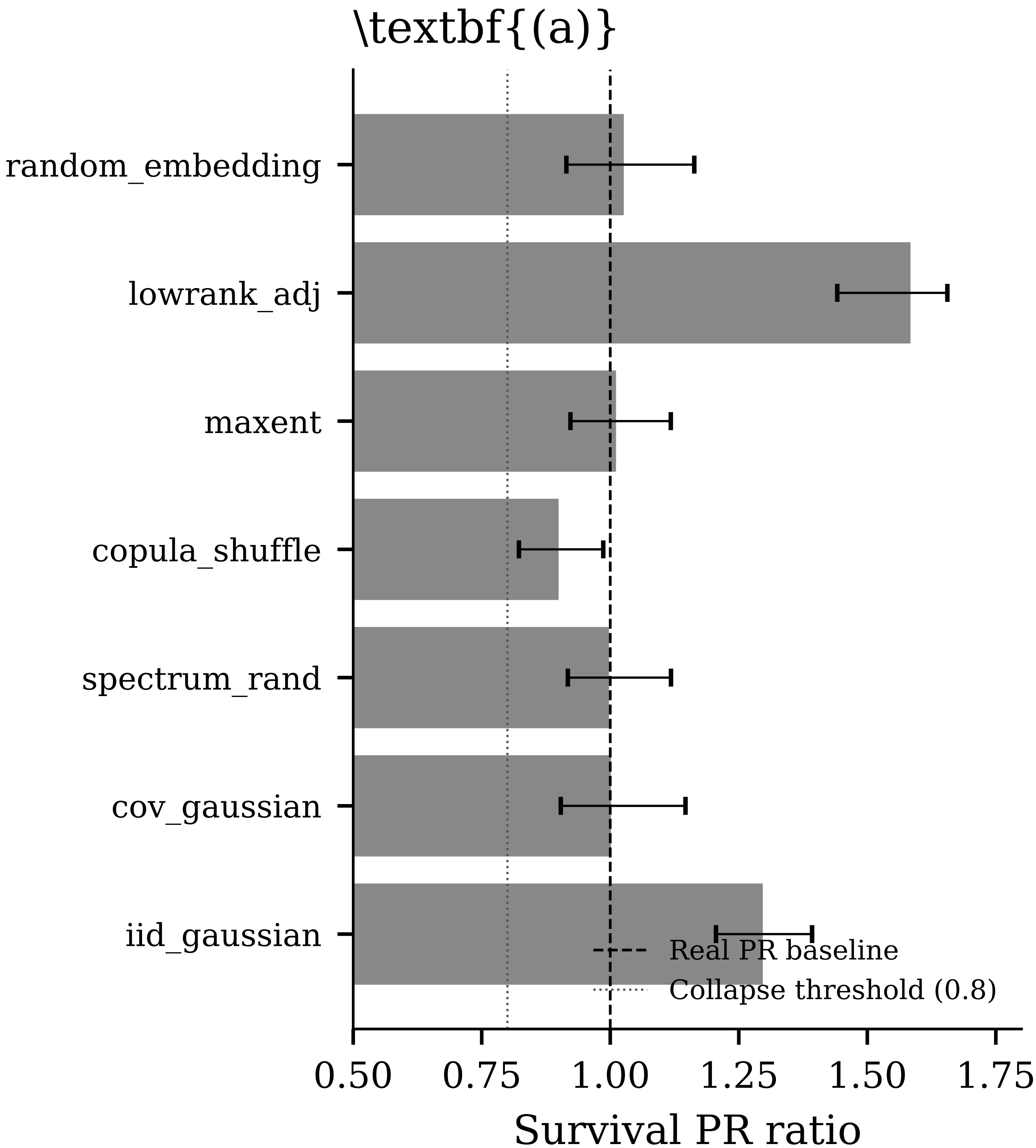

FIG. 3: Survival of manifold dimensionality under seven null-model classes [16]. Covariance-preserving Gaussian and spectral randomization reproduce participation-ratio structure nearly exactly.

D. Causal Destruction Hierarchy

Row-shuffle destruction produces the strongest collapse ($PR = 1.42$), indicating sensitivity to system-level covariance organization.

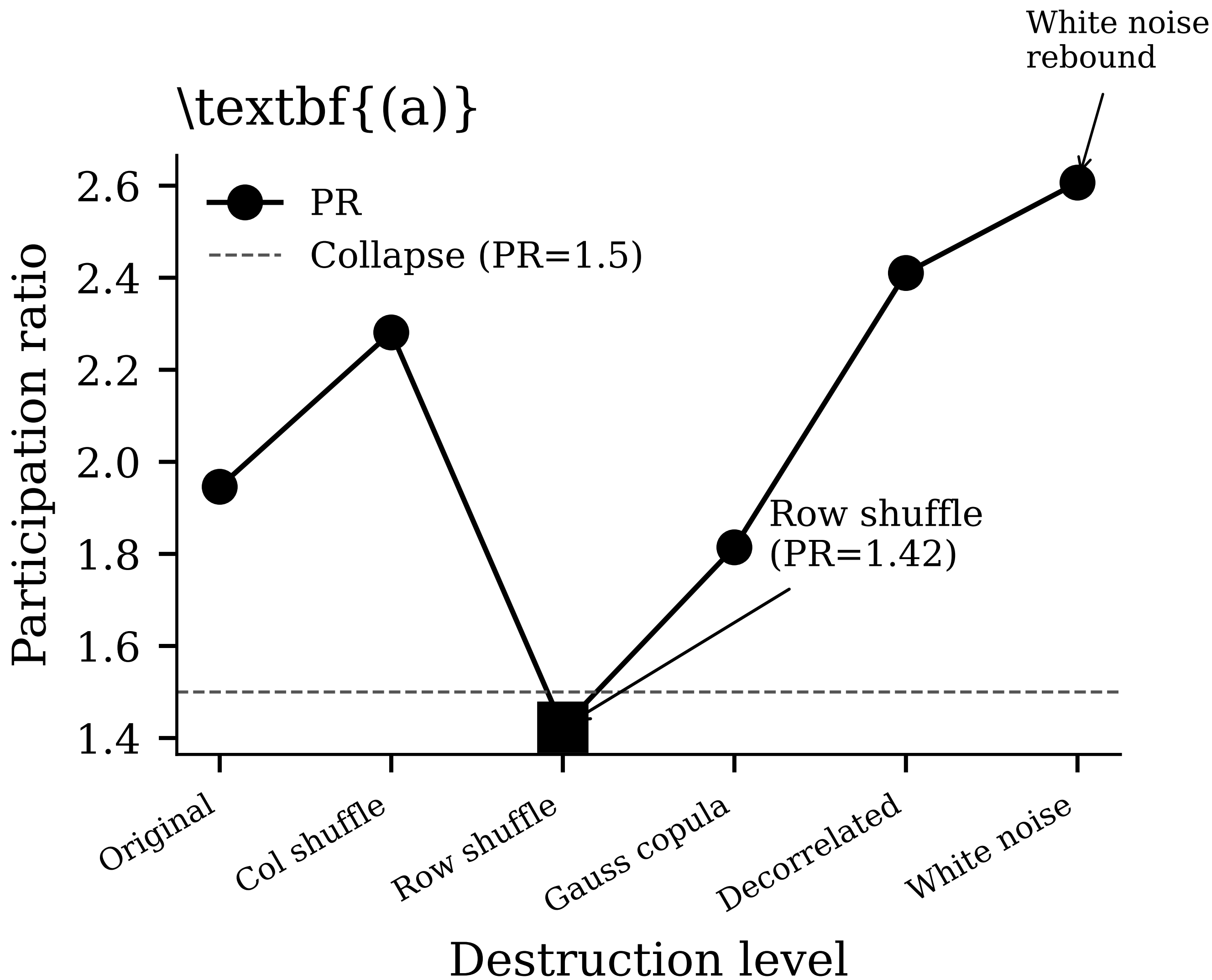

FIG. 4: Progressive destruction of system-level structure [28]. Row-shuffle collapse selectively destroys manifold persistence while white-noise projections paradoxically inflate participation ratio [29].

E. Representation Robustness

F. Decision Framework

IV. DISCUSSION

The central result is not the existence of a latent manifold itself, but the tension between dynamical persistence and statistical reproducibility [21, 33].

Several null models reproduce manifold dimensionality nearly exactly [14], indicating that covariance structure strongly constrains the geometry. However, selective collapse under row-shuffle destruction [28] and failure of adversarial low-rank nulls [15] indicate that the geometry is not purely an embedding artifact.

The work therefore supports a constrained interpretation:

1. the geometry exhibits genuine dynamical sensitivity,
2. second-order feature statistics dominate dimensionality structure,
3. manifold continuity is partial rather than universal,
4. no evidence supports universal emergence laws or ontology-level claims [34].

V. REPRODUCIBILITY

All analyses are implemented in Python with deterministic seeds. The complete pipeline is executable through a single bash entry point: `RUN_T031_PIPELINE.sh`.

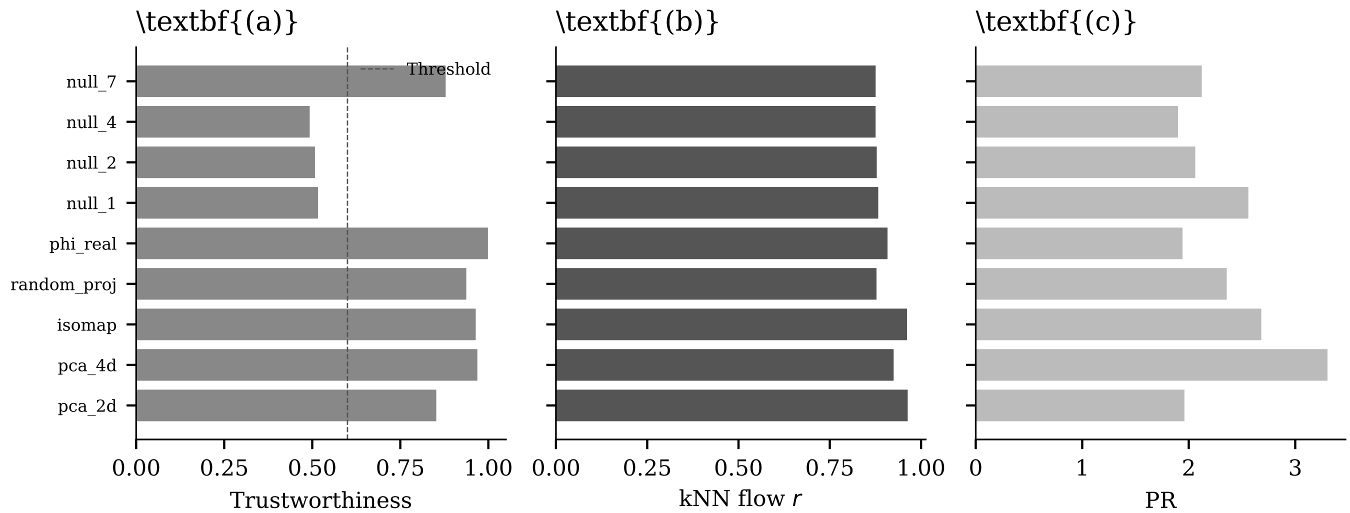

FIG. 5: Embedding robustness across PCA, Isomap [30], diffusion geometry [31], and random projection. The manifold remains partially stable across representation families.

TABLE I: Decision framework summary.

Criterion	Description	Status	Outcome
G1	Statistical null survival	Partial	Mixed
G2	Adversarial low-rank failure	Pass	Genuine
G3	Causal destruction collapse	Pass	Genuine
G4	Information geometry divergence	Pass	Genuine
G5	Representation robustness	Pass	Genuine
G6	Spectral insufficiency	Pass	Genuine
G7	Minimum collapse threshold	Pass	Genuine

Source code and figures are available through the project repository [35].

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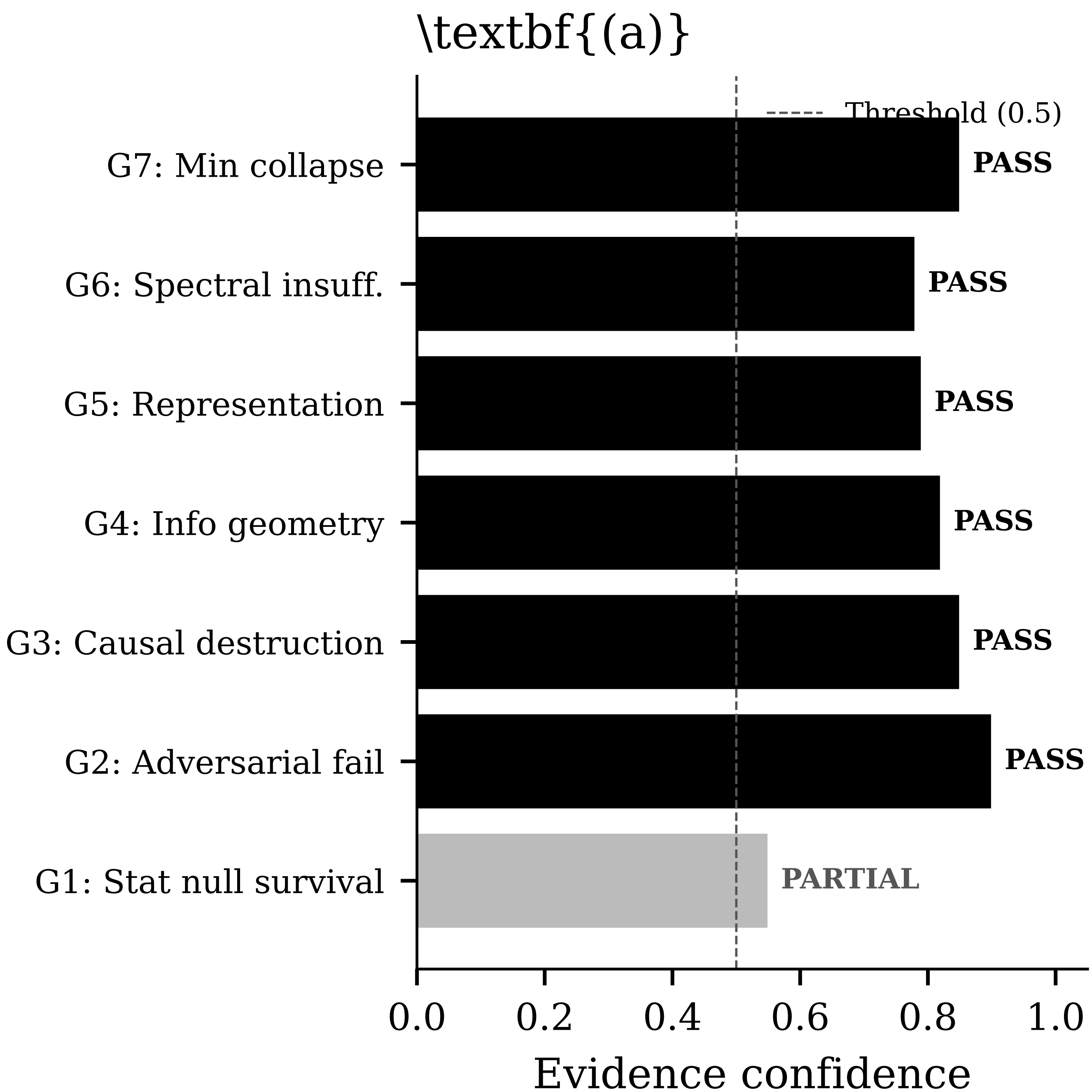

FIG. 6: Decision-framework summary [32]. Six of seven falsification criteria support a partially genuine dynamical interpretation, although substantial statistical dependence remains.

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